Residual Convolutional Neural Network Super-Resolution for Sentinel-2 Normalized Difference Vegetation Index Reconstruction in the California Central Valley

Corresponding Author: Alexander Liu
University High School
2611 E. Matoian Way, MS/UH134, Fresno, CA 93740, United States
liualexander200@gmail.com

Abstract
Groundwater overdraft has been increasingly threatening the $67 billion in annual agricultural production across the California Central Valley. However, field-scale agricultural vegetation monitoring remains limited by the spatial resolution of freely available sensors and by mixed-pixel effects at field boundaries. This study tests the hypothesis that machine learning models can improve image and NDVI reconstruction fidelity in a key agricultural region. In doing so, the work aims to advance satellite-based crop phenotyping and contribute to more precise, scalable monitoring of agricultural resilience. 15,000 HR/LR image patch pairs were generated, with 9,000 from 2018 to 2020 used for training, 3,000 from 2021 used for validation, and 3,000 from 2022 used for testing. LR patches were downsampled from HR Sentinel-2 imagery. A residual CNN trained on 9,000 HR/LR patch pairs outperformed bicubic interpolation on every evaluated metric. PSNR increased by 1.963 dB, SSIM by 0.0327, and NDVI RMSE decreased by 26.7%. Final PSNR/NDVI results come from the held-out 2022 test set. This work was benchmarked against idealized Sentinel-2 degradation and not validation against independent sub-10 m imagery. The model produced modest but measurable improvements in reconstruction quality and NDVI fidelity relative to bicubic interpolation.

Keywords: Super-resolution, NDVI, Residual CNN, California Central Valley, Remote Sensing, Image Reconstruction, Agriculture, Machine Learning, Sentinel-2, USDA NASS, CDL, PRISM

1. Introduction

The California Central Valley is a global powerhouse in agricultural production, with leading production in almonds, citrus, grapes, alfalfa, and other key crops. The agricultural sector in the Central Valley brings in $67 billion in annual revenues, supports nearly 340,000 jobs, and feeds a quarter of the United States (1). However, since the mid 1980s, groundwater has been used faster than it is being replenished (overdraft), with nearly 2 million acre-feet per year of overdraft, or 13 percent of net water use (2). This problem leads to recurrent multi-year droughts, which heavily increase groundwater and energy pressures in the southern basin, especially with the expansion of water-intensive tree crops like almonds (3). Mixed pixels and field heterogeneity currently limit the ability of satellite-based monitoring systems to reconstruct vegetation patterns at fine spatial scales. Improving the spatial and temporal resolution of agricultural monitoring is therefore critical for understanding crop-specific growth dynamics and long-term resilience.

Satellite remote sensing has emerged as a central tool for large-scale agricultural monitoring, offering repeated, cost-effective observations of vegetation condition, canopy structure, and phenology across entire regions. Compared to airborne or UAV-based (unmanned aerial vehicle) platforms, satellite systems offer high scalability and temporal consistency, making them well suited for multi-year regional analyses. However, the utility of satellite phenotyping remains constrained by spatial resolution limits, mixed-pixel effects, cloud contamination, and the difficulty of resolving fine-scale variability within and between crop fields (4). These limitations are especially consequential in the Central Valley, which has diverse crop types, often varying within fields.

Recent progress in artificial intelligence (AI) has substantially expanded satellite-image analysis capabilities. Deep learning methods, including convolutional neural networks and transformer-based architectures have improved feature extraction, classification accuracy, and temporal modeling of satellite imagery across environmental monitoring applications (5). Specifically, machine learning models have been used on data from satellites, such as from the Zhuhai-1 hyperspectral satellite, to enhance the assessment of plant health and field coverage (6). Moreover, a multi-modal deep learning framework that combines Landsat imagery, climate variables, evapotranspiration, and soil data to forecast crop yields across California was developed by researchers (7). While this work demonstrates the predictive power of fused satellite-climate datasets over long time horizons, it's limited in the way that its resolution is only county-level, which reveals its inability to resolve intra-county and field-scale variability in crop response. While prior studies have used satellite and climate data for crop monitoring, fewer have tested whether learned super-resolution (SR) improves NDVI reconstruction relative to bicubic interpolation on crop-filtered imagery.

This study asks whether a lightweight residual CNN can reconstruct native-resolution Sentinel-2 reflectance and NDVI more accurately than bicubic interpolation on crop-filtered agricultural imagery. SR is a computational technique that reconstructs higher-spatial-resolution imagery from lower-resolution inputs, often using machine learning to infer fine-scale spatial structure.

This study tests the hypothesis that AI-enhanced satellite super-resolution will improve image and NDVI reconstruction fidelity on crop-filtered agricultural imagery (alfalfa, almonds, and grapes) across multi-year higher and lower precipitation periods. The experiment tested whether learned reconstruction could better recover native-resolution spatial and spectral structure than bicubic interpolation after controlled 4x degradation. In doing so, the work aims to advance satellite-based crop phenotyping and contribute to more precise, scalable monitoring of agricultural resilience in water-limited regions. Granted, this study does not propose a novel super-resolution architecture; this study evaluates whether a lightweight residual CNN can improve both image reconstruction and vegetation-index fidelity over bicubic interpolation in a controlled agricultural Sentinel-2 benchmark.The study compared native-resolution and super-resolved NDVI patterns and reconstruction quality metrics such as Peak Signal-to-noise Ratio (PSNR), Structural Similarity Index Measure (SSIM), Mean Absolute Error (MAE), and Mean Square Error (MSE).

2. Methods and Materials

2.1 Datasets and Study Area

Datasets used: Copernicus Sentinel-2 Surface Reflectance Harmonized for Sentinel Data (8), USDA NASS Cropland Data Layer (9), and PRISM ANm for monthly PRISM climate data data (10) from January 2018 to December 2022. Earth Engine IDs used were COPERNICUS/S2_SR_HARMONIZED, USDA/NASS/CDL, OREGONSTATE/PRISM/ANm respectively. Model training was performed in PyTorch (11) using CUDA acceleration, with a random seed of 42 for reproducibility. Datasets were accessed August 8, 2026.

The area of interest (AOI) is defined geographically by the bounding box: latitude: 38.0°N to 38.5°N, & longitude: 121.0°W to 122°W. The AOI was chosen for its presence of key crop classes such as grape, alfalfa, and almond. Sentinel-2 surface reflectance imagery from Google Earth Engine was used over a key crop growing region: the Sacramento to San Joaquin Delta region from January 2018 to December 2022. The multi-year approach was taken in order to assess crop growth patterns in response to different hydrological conditions.

2.2 High/Low Precipitation Year Labeling

Determining high or low precipitation years was done by using PRISM precipitation data over the AOI to compute the yearly annual precipitation and comparing it to the (2018 to 2022) mean. Years with annual precipitation levels below the multi-year mean were labeled low precipitation years, while years above the multi-year mean were labeled high precipitation years. This is a relative grouping, not a climatological drought classification.

Table 1. Precipitation Labelling Summary

Year	Annual precipitation (mm)	Anomaly from study-period mean (mm)	Precipitation label
2018	436.5	51.7	Higher Precipitation
2019	575.9	191.1	Higher Precipitation
2020	171.5	-213.3	Lower Precipitation
2021	466.3	81.5	Higher Precipitation
2022	273.9	-111	Lower Precipitation

2.3 NDVI Processing

Sentinel-2 scenes were filtered so that the cloudy pixel percentage must be less than or equal to 60% in each scene. Cloud masking was then applied using the Scene Classification Layer (SCL) by retaining classes 4, 5, 6, 7 (vegetation/bare soil/water/unclassified respectively). SCL classes 5, and 6 were retained because they do not present atmospheric obstructions, and they include important contextual information as they occur nominally in crop patches. Class 7 contains low-probability cloud/unclassified pixels and was retained for contextual coverage, although residual contamination is therefore possible. All other classes (clouds, snow, no data, and other obstructions) were excluded as they presented atmospheric obstructions where observed reflectance does not represent the actual land surface. Crop masks were year-specific CDL layers; a matching CDL layer was pulled for each year.

Vegetation greenness was quantified using the normalized difference vegetation index (NDVI). NDVI was computed per image as seen in (eq1).

$$NDVI = \frac{B8-B4}{B8+B4} \quad \text{(eq1)}$$

B8 is the near infrared band, and B4 is the red band (12). Sentinel-2 scenes were then processed individually. For each scene, mean NDVI was calculated over pixels assigned to the corresponding crop class within the AOI. Crop pixels were identified using the United States Department of Agriculture (USDA) Cropland Data Layer (CDL) for each year and masking to class codes for Alfalfa, Almond, Citrus, and Grape (36, 75, 72, 69 respectively). Mean NDVI time series were computed by reducing NDVI over crop-masked pixels within the AOI at 60 m scale. High/low precipitation year labels were derived from the process described in the High/Low Precipitation Year labeling section. A baseline NDVI was built from 538 scene-observations of Alfalfa, 537 of Almond, 366 of citrus, and 538 of grapes from 2018–2022.

2.4 Super-Resolution Dataset Construction

For each year, a cloud-masked June–September median Sentinel-2 composite was generated. HR patches were extracted from these growing-season composites. Four Sentinel-2 bands were used: B4, B3, B2, and B8, corresponding to red, green, blue, and near-infrared (NiR) reflectance. HR 128x128 pixel (1.28km by 1.28 km) Sentinel-2 patches were downsampled by a scale of 4x to 32x32 pixels (LR) using area-averaging and then reconstructed back to 128x128 using bicubic interpolation. A crop coverage threshold of 0.30 was used, meaning that at least 30% of CDL pixels within a candidate patch had to belong collectively to the set of target crop classes.No retained SR patch had citrus as its dominant target-crop label. Citrus returned zero patches for all five years, so it was excluded.

In total, 15,000 HR/LR pair patches were generated (3,000 patches per year, across 5 years). Patches were sampled during growing seasons. The 2018, 2019, and 2020 patches were assigned as training patches, 2021 as validation, and 2022 as testing. This chronological hold out strategy was implemented in order to avoid same-composite leakage between training, validation, and test data. Because of the small AOI, combined with a large patch size (1.28 km), a random training/validation/test split meant that a model could potentially "recognize" a field it already saw. This strategy tests temporal generalization within the same region rather than complete spatial independence, however. A patch split map was generated in order to visualize patch distribution (figure 1.)

Patches were verified as valid through a series of validity checks before being retained in the training set. During generation, HR patches were checked if they contained any non-finite pixel values, if more than 5% of its pixels were exactly zero across four bands (which indicated missing/masked data), or if its pixel-value standard deviation fell below $1x10^{-6}$ (which indicated a blank or degenerate patch. If any of these criteria were met, the patch was rejected. All retained pairs were then checked for the correct dimensions, correct band order (B4, B3, B2, B8), that all pixel values fell within the reflectance range of [0,1] ($\pm 1x10^{-6}$ tolerance), and confirmed for a second time that they contained only finite-pixel values. The LR array was also checked to see if it matched a 4x area-average downsampling of the corresponding HR array within a numerical tolerance of $1x10^{-4}$. A patch was only classified as valid if it passed every check. An audit was performed to check coordinate pairs for uniqueness.

2.5 Model Training

The dataset was loaded via a custom PyTorch (11) dataset class that reads each LR/HR pair and rescales pixel values to [0,1], then returns paired tensors. Training and Validation sets were wrapped in Pytorch dataloaders with a batch size of 16. The test set was evaluated patch-by-patch using batch size 1 so that each of the 3,000 test patches received an individual metric value in order to confirm every metric was a genuine per-patch value rather than a batch-size-dependent average.

The model was trained by minimizing a composite loss consisting of L1 reconstruction loss plus a gradient-difference loss (0.05 weight) that penalizes discrepancies in horizontal and vertical image gradients between prediction and target.

The upsampled image described in the previous section was then refined with a residual CNN. The refinement network consisted of an entry convolution (5x5 kernel, 4 → 64 channels, padding 2, ReLU activation), 6 residual blocks (3x3 convolutions, 64 → 64 channels, padding 1, ReLU activation after the first convolution and a residual connection followed by a second ReLU after the block sum), and an exit convolution (3x3 kernel, 64 → 4 channels, padding 1). The input and output both have 4 channels (B4, B3, B2, B8), and the model has approx. 452,000 trainable parameters: 6,464 in the entry layer, 443,136 across the 6 residual blocks, and 2,308 in the exit layer.

The model was optimized with the Adam optimizer at a fixed learning rate of $5x10^{-4}$ over 40 epochs with a patience of 8 epochs without validation-loss improvement. The checkpoint with the lowest validation loss was retained; in the final run, this occurred at epoch 39.

3. Results

3.1 Baseline Crop Specific NDVI Patterns

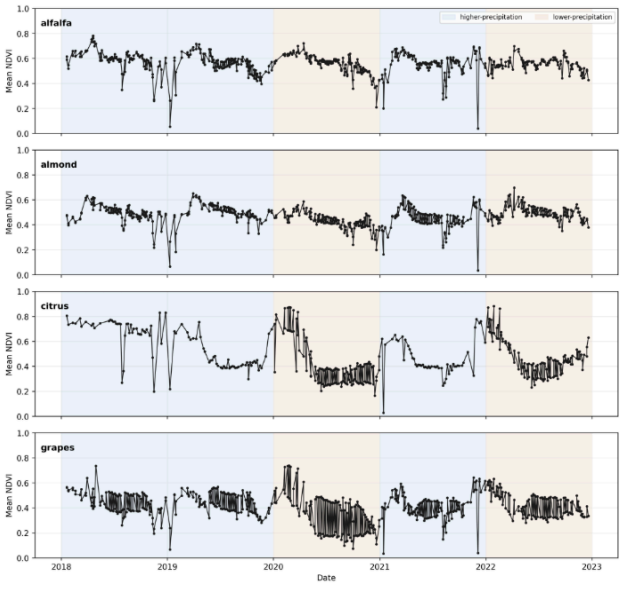


Figure 1: Baseline crop-specific NDVI time series, 2018-2022

As shown in figure 1., the alfalfa NDVI curve exhibits a highly dynamic, multi-cut seasonal signature, with rapid oscillations between approximately 0.04 and 0.80 reflecting the repeated harvest-and-regrowth cycles characteristic of alfalfa production throughout the growing season. During higher-precipitation years (2018, 2019, 2021), NDVI peaks reach as high as 0.79, with consistently elevated inter-cut recovery values in the 0.60 to 0.65 range. During lower-precipitation years (2020, 2022), the overall NDVI envelope trends downward, with mean values settling in the 0.43–0.55 range and more frequent deep troughs dropping below 0.25.

The almond NDVI curve exhibits a clear seasonal pattern driven by the species' deciduous phenology, with values rising from a dormant season baseline near 0.40 to 0.48 in January through a spring canopy flush peaking around 0.60 to 0.65 by mid-summer, before declining again through autumn senescence. During higher-precipitation years, spring green-up is robust and early, with peak NDVI values reaching 0.63 to 0.65 and a sustained high-greenness period

extending through summer. In lower-precipitation years, a pronounced and sustained depression of NDVI is evident, with values declining from a shared crossing point near 0.50 in early 2020 (figure 1.).

The citrus NDVI curve exhibits a relatively stable, evergreen seasonal signature, with mean NDVI values ranging from approximately 0.72 to 0.76 in early 2018, reflecting the persistent canopy cover characteristic of citrus orchards year-round. NDVI shows a gradual suppression during summer and early fall months, with values declining toward 0.35 to 0.45. High-precipitation labeled years correspond to elevated and more variable NDVI signals, particularly during winter and spring flush periods.

The grape NDVI curve follows a predictable annual pattern where NDVI starts low in winter, rises to a single peak in summer as the canopy fills out, then falls again in autumn as leaves drop. It rises from a dormant-season baseline of approximately 0.40 to 0.55 in winter through a canopy expansion peak in mid-summer, before declining sharply through autumn harvest and leaf drop. During higher-precipitation years, NDVI maintains a relatively stable and elevated mid-season envelope in the 0.45 to 0.60 range, with consistent seasonal cycling and peak values reaching approximately 0.60 to 0.65. In lower-precipitation years, the NDVI signal undergoes a dramatic collapse, dropping to values as low as 0.10 to 0.20 through much of 2020 before partially recovering in 2021.

All four crops showed a decline in mean NDVI during lower-precipitation years relative to high precipitation years. Citrus showed a decline of -17.5%; grape, -10.9%; almond, -4.0%; and alfalfa at -2.3% (table 1.).

Table 1. Baseline NDVI Summary

Crop	Higher-precipitation NDVI mean ± SD	Lower-precipitation NDVI mean ± SD	Difference	% change	n higher	n lower
Alfalfa	0.567 ± 0.094	0.554 ± 0.066	-0.013	-2.3	310	228
Almond	0.478 ± 0.085	0.459 ± 0.068	-0.019	-4	309	228
Citrus	0.532 ± 0.164	0.439 ± 0.164	-0.093	-17.5	148	218
Grapes	0.429 ± 0.095	0.382 ± 0.134	-0.047	-10.9	310	228

Note: Values shown here are scene-level descriptive summaries. The scene counts should not be interpreted as independent biological replicates because repeated satellite observations from the

same region are temporally correlated. These values are therefore presented descriptively rather than as independent-sample statistical inference.

3.2 Model Performance

The trained model outperformed the bicubic baseline on every metric. Metrics were calculated on 3,000 held-out test patches from 2022. Training converged smoothly over 40 epochs, with validation PSNR rising from 36.595 to 38.558 dB, SSIM increasing by 0.0327, NDVI RMSE decreasing by -0.0178, and MAE and MSE decreasing by -0.00197 and -8.26x10^{-5} respectively at the best epoch (39) (Table 2.). Validation loss decreased monotonically with minor fluctuation (figure 2.).

Table 2: Model Performance

Metric	Bicubic (mean ± sd)	Model (mean ± sd)	Δ (Model − Bicubic)
MAE	0.00873 ± 0.00210	0.00676 ± 0.00188	−0.00197
MSE	0.000242 ± 0.000112	0.000159 ± 0.000088	−0.0000826
PSNR (dB)	36.60 ± 1.95	38.56 ± 2.23	1.96
SSIM	0.9072 ± 0.0271	0.9399 ± 0.0230	0.0327
NDVI RMSE	0.0668 ± 0.0232	0.0490 ± 0.0148	−0.0178

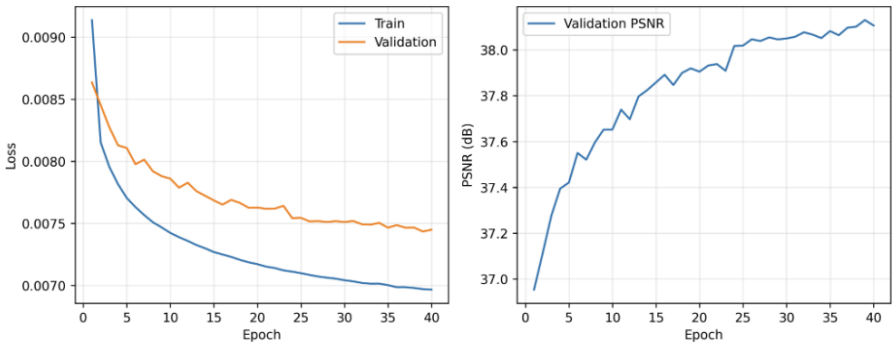


Figure 2: Residual CNN Training Curves

As seen in figure 3 & 4., visual comparison of LR input, bicubic upsampling, model prediction, and HR target show the model outputs appear to preserve sharper field boundaries than bicubic. NDVI maps show the model's field boundary sharpness closely tracking HR target's parcel structure. Model NDVI error portrays residual error concentrated along field boundaries.

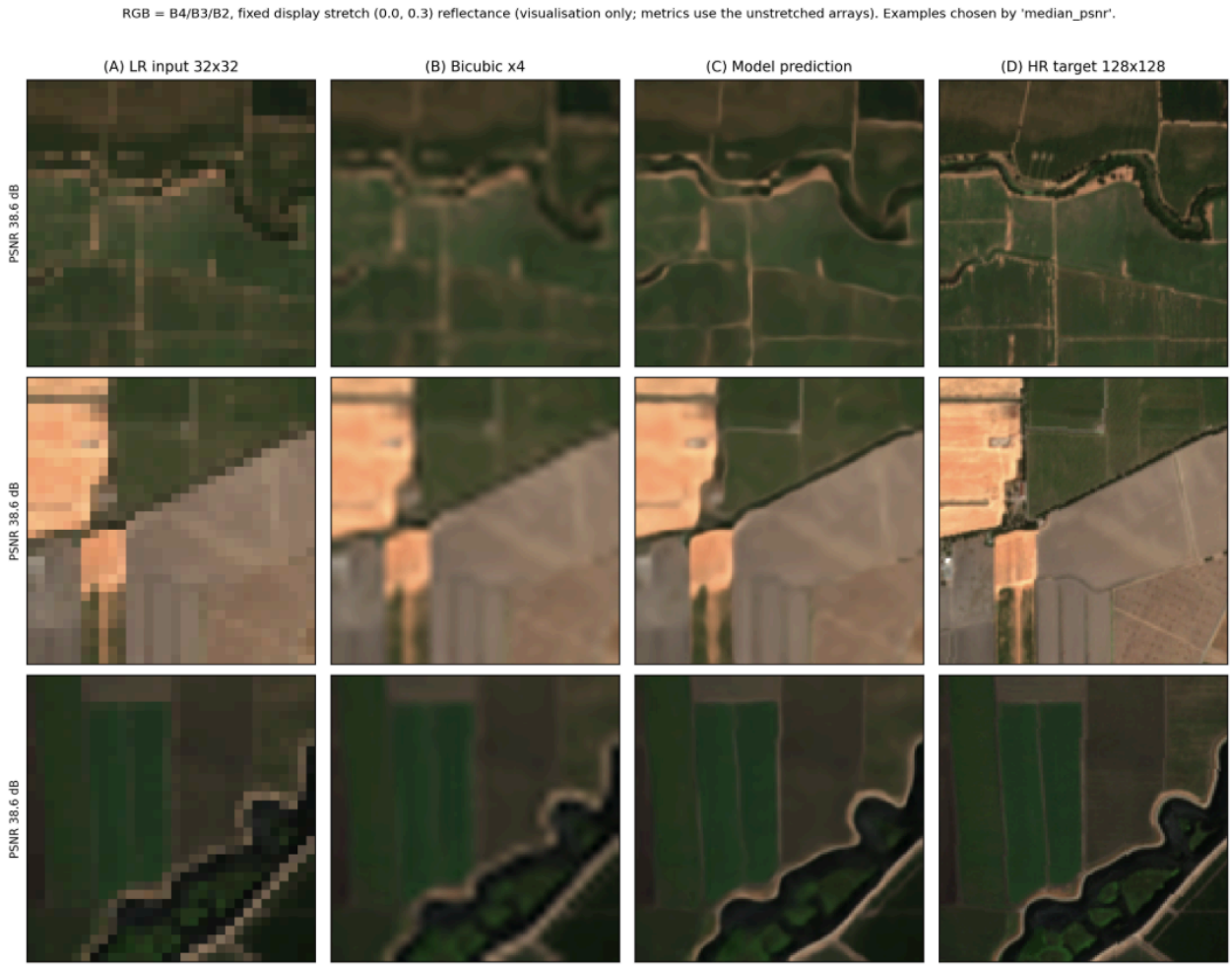


Figure 3: Visual comparison of super-resolution outputs on held-out test patches

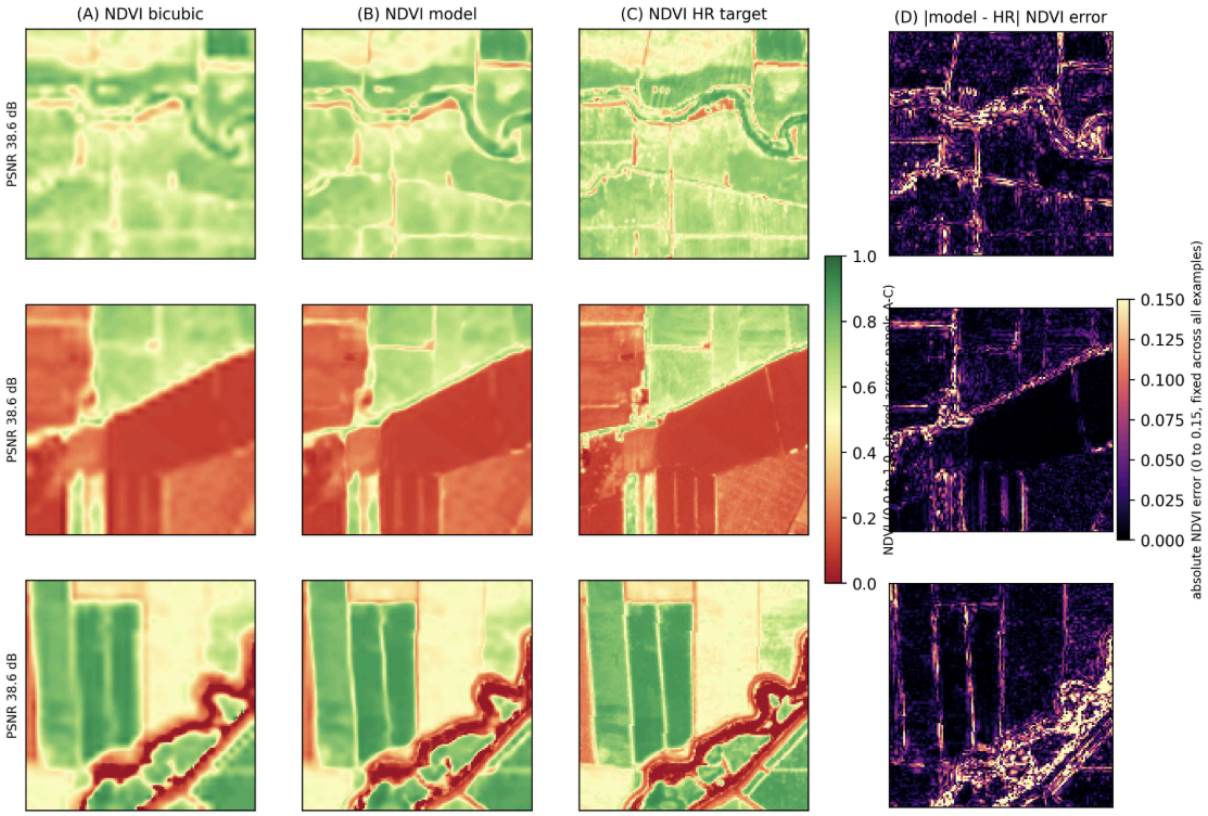


Figure 4: NDVI reconstruction and error for the patches in Figure 3.

4. Discussion

4.1 Baseline NDVI Analysis

Alfalfa showed lower mean NDVI and more frequent low-NDVI excursions during lower-precipitation years, a pattern that may be consistent with reduced canopy recovery between harvest cycles. The high intra-annual variability and tight spatial intermixing of harvest stages across fields motivate the use of super-resolution to distinguish individual field boundaries and cutting cycles at finer spatial scales than 10 m composites allow.

Compared to alfalfa, almond's lower-precipitation signal is expressed as a slower, multi-season decline rather than acute within-season troughs, reflecting the buffering capacity of deep tree root systems that eventually become overwhelmed under sustained deficit irrigation: a pattern that motivates super-resolution analysis to resolve individual orchard blocks, and the result motivates future work testing whether higher-fidelity reconstruction can improve orchard-block-scale vegetation monitoring.

Citrus NDVI shows a gradual suppression during summer and early fall months, with values declining toward 0.35 to 0.45, suggesting moisture stress reducing canopy greenness even in this perennial crop. Compared to almond, citrus maintains a higher baseline NDVI through winter

months but shows comparably sharp lower-precipitation driven declines in peak summer, motivating the use of super-resolution to better improve delineation of orchard-scale vegetation patterns and capture within-orchard stress variability at finer spatial scales. Please note that citrus is only included in the baseline analysis; it is excluded from the SR results.

The Grape NDVI is indicative of well-supported vine canopy development under adequate moisture conditions. The lower-precipitation depression in grapes is among the most acute observed across all four crops in this study, with the 2020 growing season showing a near-complete loss of the expected seasonal NDVI rise. This pattern is distinctly more severe than the gradual multi-season decline seen in almonds and the cutting-cycle suppression seen in alfalfa. The high precipitation years show notable high-frequency variability in the orange trace, likely reflecting a mix of grape management practices (canopy trimming, cover crop presence) and field heterogeneity across the AOI, underscoring the need for super-resolution imagery to better preserve field-boundary and within-parcel spatial variation relative to bicubic interpolation.

4.2 Image Reconstruction

The residual CNN beat bicubic interpolation across the board on the held-out test set, but by small margins. PSNR rose by ~1.963 dB, a difference that is measurable. The SSIM improved by 0.0327. The MAE and MSE reductions are proportionally larger (-22.5% and -34.2% respectively). Both bicubic and the model are already producing very low error (MAE ≈ 0.009 in reflectance). The learned residual correction is doing real, consistent work on top of bicubic upsampling, but it is a refinement of an already-good baseline rather than a transformative jump in image quality. These results are expected for a lightweight 6-block residual CNN trained on 9,000 patches from a single region dataset.

4.3 NDVI Reconstruction

NDVI RMSE improved ~26.7% (0.0668 to 0.0490). This is a meaningful result because NDVI is a nonlinear ratio of the red and NIR bands, so pixel-level errors in each band do not translate into NDVI error. This indicates that the learned residual correction improved preservation of the red/NIR relationship relative to bicubic interpolation. This matters for downstream applications such as lower-precipitation stress or crop-health monitoring, where the vegetation index is the quantity of interest, rather than raw pixel fidelity. However, NDVI is a nonlinear ratio, so red and NiR-band errors do not translate linearly into NDVI error.

4.4 Limitations

While the model's improvement over bicubic was consistent, the gains were incremental rather than transformative for several reasons. The biggest factor in model performance was the limited geographic, temporal, and source-image diversity. Patches consisted of only 3 crop types—with grapes dominating the composition—and only 1 region over 5 years. Moreover, Central Valley cropland varies substantially in field boundaries, cover crops, irrigation patterns, and canopy structure, even within single crop classes. Furthermore, residual contamination from retaining unclassified pixels may have impacted model performance. A single model has to learn mapping

across this diversity, which in combination with the limited training data, limits how accurate and sharp any single reconstruction can be. Most importantly, the LR images were synthetically generated by area-averaging the same real Sentinel-2 HR image, rather than being paired with an independent, genuine LR sensor. This means that both the HR and LR imagery share identical atmospheric conditions and geometry, meaning that the model is learning to reconstruct an image that is already known, rather than a true-cross sensor resolution gap using its own point-spread function, noise, and registration error.The chronological split prevented patches from the same annual composite from appearing across training, validation, and test sets; however, the same geographic fields may recur across different years, so the test assesses temporal generalization within one region rather than complete spatial independence.

Together, these factors suggest that the 1.963 dB PSNR gain and the 26.7% NDVI RMSE reduction represent a proof-of-concept estimate under idealized degradation conditions of the potential of what a scale-aware residual SR model can offer for this application. Performance against independent higher-resolution observations remains unknown and may be lower than the performance measured under this idealized degradation benchmark.

5. Conclusion and Future Work

This study demonstrated that a lightweight, residual CNN can super-resolve Sentinel-2 imagery in a way that measurably improves both raw image quality and the fidelity of NDVI. The model improved reconstruction and NDVI fidelity relative to bicubic interpolation on an idealized Sentinel-2 degradation benchmark. At the same time, these gains should be read in proportion to the constraints of the current study design. Future work should expand the AOI, increase the number of years studied, include additional crop classes, and expand the training set. Moreover, ground truth imagery should be used to test whether the model's gains hold against real sensor degradation, rather than synthetically generated degradation. The results should be understood as a proof-of-concept. Residual SR can add real value to NDVI monitoring, however the model in this study should not be seen as a valid operational tool ready for deployment on independently sensed imagery.

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Generative AI tools were used during development for code debugging suggestions and editorial feedback. The author reviewed, executed, validated, and takes full responsibility for the final code, analyses, interpretations, and manuscript.

7. Data and Code Availability

All analysis code used in this study has been deposited on Github under the MIT License at https://github.com/AlexLiu24/CentralValleySR.git

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9. Conflict of Interest

The author declares that there are no conflicts of interest related to this work.

10. References

1. Escriva-Bou, A., Hanak, E., Cole, S., & Medellín-Azuara, J. (2023). *Policy brief: The future of agriculture in the San Joaquin Valley*. Public Policy Institute of California. https://www.ppic.org/publication/policy-brief-the-future-of-agriculture-in-the-san-joaquin-valley/
2. Hanak, E., Lund, J., Arnold, B., Escriva-Bou, A., Gray, B., Green, S., Harter, T., Howitt, R., MacEwan, D., Medellín-Azuara, J., Moyle, P., & Seavy, N. (2017). Water Stress and a Changing San Joaquin Valley. *Public Policy Institute of California*
3. Alam, S., Gebremichael, M., & Li, R. (2019). Remote Sensing-Based Assessment of the Crop, Energy and Water Nexus in the Central Valley, California. Remote Sensing, 11(14), 1701. https://doi.org/10.3390/rs11141701
4. Zhang, C., Marzougui, A., & Sankaran, S. (2020). High-resolution satellite imagery applications in crop phenotyping: An overview. *Computers and Electronics in Agriculture*, *175*, 105584. https://doi.org/10.1016/j.compag.2020.105584
5. Alzubaidi, L., Zhang, J., Humaidi, A. J., Al-Dujaili, A., Duan, Y., Al-Shamma, O., Santamaría, J., Fadhel, M. A., Al-Amidie, M., & Farhan, L. (2021). Review of deep learning: Concepts, CNN architectures, challenges, applications, future directions. *Journal of Big Data*, *8*(1), 53. https://doi.org/10.1186/s40537-021-00444-8
6. Ojo, O. L., Ajiboye, Y., Afolalu, A. S., Morebise, A. T., Omoyajowo, I. M., Abe, O. E., Olumodimu, O., Adeyeye, D. S., Agboola, O. M., & Olawunmi, O. (2024). AI Applications in Satellite Image Processing: Enhancing Earth Observation and Environmental Monitoring. 2024 IEEE 5th International Conference on Electro-Computing Technologies for Humanity (NIGERCON), 1–5. https://doi.org/10.1109/NIGERCON62786.2024.10926999
7. Kamangir, H., Hajiesmaeeli, M., & Earles, M. (2025). California Crop Yield Benchmark: Combining Satellite Image, Climate, Evapotranspiration, and Soil Data Layers for County-Level Yield Forecasting of Over 70 Crops (No. arXiv:2506.10228). arXiv. https://doi.org/10.48550/arXiv.2506.10228
8. European Union/ESA/Copernicus. (2018–2022). *Sentinel-2 MSI Level-2A Surface Reflectance Data*. Copernicus Data Space Ecosystem. Retrieved August 8, 2026, from https://dataspace.copernicus.eu
9. USDA NASS, 2026, *2018–2022 Cropland Data Layer*: USDA NASS, USDA NASS Marketing and Information Services Office, Washington, D.C. Available at: https://croplandcros.scinet.usda.gov/
10. PRISM Group, Oregon State University. (2018–2022). *PRISM monthly spatial climate dataset [ANm]*. Retrieved August 8, 2026, from https://prism.oregonstate.edu
11. Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., Killeen, T., Lin, Z., Gimelshein, N., Antiga, L., Desmaison, A., Kopf, A., Yang, E., DeVito, Z., Raison, M., Tejani, A., Chilamkurthy, S., Steiner, B., Fang, L., Bai, J., & Chintala, S. (2019). PyTorch: An imperative style, high-performance deep learning library. In H. Wallach, H. Larochelle, A. Beygelzimer, F. d'Alché-Buc, E. Fox, & R. Garnett (Eds.), *Advances in*

Neural Information Processing Systems 32 (pp. 8024–8035). Curran Associates, Inc. neurips.cc

12. Huang, S., Tang, L., Hupy, J.P. *et al.* A commentary review on the use of normalized difference vegetation index (NDVI) in the era of popular remote sensing. *J. For. Res.* 32, 1–6 (2021). https://doi.org/10.1007/s11676-020-01155-1